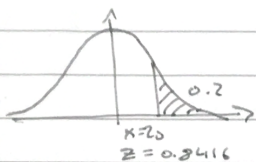


Missing μ or σ in Normal.

" $X \sim N(\mu, 3^2)$, Given $P(X > 20) = 0.2$, find μ "



① Using $Z \sim N(0, 1)$
 $z = 0.8416$

② as $\frac{x - \mu}{\sigma} = z$

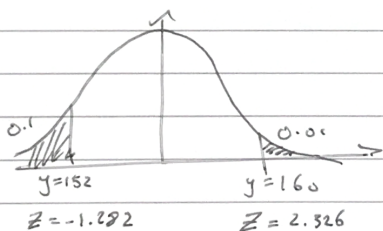
use this equation

$$\begin{aligned} \mu &= x - z\sigma \\ &= 20 - 0.8416 \times 3 \\ &= \underline{\underline{17.48}} \end{aligned}$$

Both unknown:

→ Use sim. eq.

" $Y \sim N(\mu, \sigma^2)$, give $P(Y < 160) = 0.99$ & $P(Y > 152) = 0.90$
Find σ & μ "



$$-1.282 = \frac{152 - \mu}{\sigma}$$

$$\sigma = \frac{152 - \mu}{-1.282}$$

$$2.326 = \frac{160 - \mu}{\sigma}$$

$$\therefore \sigma = \frac{160 - \mu}{2.326}$$

$$\Rightarrow \sigma = 2.247$$

$$\Rightarrow \mu = 154.8$$